ALGEBRA 1 - UNIT 1

Section 1.4
Expressions and Structure

Looking inside expressions to see how they are built.

> **Mathematical idea:** An expression is not just a string of symbols. Its structure can show parts, meaning, and smarter ways to rewrite it.

Today we will identify parts of expressions, combine like terms, use the distributive property, and explain why equivalent forms can reveal different information.

Learning Target

Students will interpret and rewrite algebraic expressions by identifying structure, meaning, and equivalent forms.

I can statements

- I can identify parts of an expression.
- I can simplify expressions.
- I can write equivalent expressions.
- I can explain what an expression means in context.
- I can use structure to choose an efficient rewrite.

What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: What parts of each expression do you recognize? Which forms look connected? Which form seems easiest to use for substitution? Which form seems best for explaining the story?

Future Task

A student says the four cards below are all connected, but each one reveals something different about the same situation.

Card A

$$3(x+4)+2x$$

Shows a group of $x+4$ repeated three times, plus another $2x$.

Card B

$$3x+12+2x$$

Shows the parts after the first group has been distributed.

Card C

$$5x+12$$

Shows the simplified total after like terms are combined.

Card D

$$x+x+x+12+x+x$$

Shows the expression as repeated pieces.

1. Annotate what each card seems to show.
2. Circle two cards that you think are easiest to compare.
3. Underline one form that seems most useful for mental math.
4. Write one question you would need to answer before proving all four cards are equivalent.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

An algebraic expression is a math phrase. It can have numbers, variables, operation symbols, and grouping symbols. Expressions do not have an equals sign because they do not make a complete statement yet.

Structure means how something is built. In an expression, structure helps us see the parts. For example, in $4x+7$, the term $4x$ changes when x changes, but the 7 stays the same. In $3(x+2)$, the parentheses show a group being repeated three times.

Sometimes two expressions look different but have the same value for every input. These are called equivalent expressions. We can rewrite expressions by combining like terms or using the distributive property.

The goal is not just to "get an answer." The goal is to understand what the expression is showing so we can choose a helpful form and explain why it works.

Stop and Discuss

1. What does it mean to look at the structure of an expression?
2. Why might two expressions look different but still be equivalent?
3. Why is rewriting an expression sometimes more helpful than evaluating it right away?

Expressions Have Parts That Do Different Jobs

Before rewriting an expression, we need to notice its parts. Terms are separated by addition or subtraction. A coefficient is the number multiplying a variable. A constant is a number without a variable.

Term: a part of an expression separated by addition or subtraction.

Coefficient: the number multiplying a variable.

Constant: a number that does not have a variable attached.

Example 1

Identify the terms, coefficients, and constant in the expression.

$$6x+9-2x+5$$

terms **coefficients** **constant terms** **like terms**

Then explain which parts could be combined and why.

You Try It 1

Identify the terms, coefficients, and constant terms in $8n-3+4n+10$. Then circle the like terms in your work.

Stop and Discuss

1. Why is $6x$ a term but 6 is the coefficient?
2. How can recognizing like terms help before you simplify?

Like Terms Can Be Combined

Like terms have the same variable part. Combining like terms means adding or subtracting the coefficients while keeping the variable part the same.

Like terms: terms with matching variable parts, such as $5x$ and $2x$, or 9 and -4.

Example 2

Simplify the expression and explain which terms were combined.

$$5x+3+2x-1$$

Use the structure to group variable terms with variable terms and constants with constants.

You Try It 2

Simplify $7a-4+3a+9$. In one sentence, explain why the variable terms can combine but the variable terms and constants should not combine together.

Parentheses Show a Group

The distributive property lets us rewrite a group being multiplied. It does not create a new relationship. It creates an equivalent expression that shows the same value in a different form.

Distributive property: $a(b+c)=ab+ac$.

Example 3

Use the area model to explain why $4(x+6)$ is equivalent to $4x+24$.

height 4	
x **by** 4 **gives** $4x$	6 **by** 4 **gives** 24

Then write a sentence connecting the picture to the symbols.

You Try It 3

Rewrite $3(y+5)$ using the distributive property. Then explain what each new term came from.

Stop and Discuss

1. Why must the outside number multiply every term inside the parentheses?
2. How does an area model help show equivalent expressions instead of just memorizing a rule?

Different Forms Can Be Useful for Different Reasons

An expression can be written in more than one equivalent form. One form may show the story better. Another form may make substitution or mental math easier.

Example 4

A snack box has 2 bags of pretzels and 2 juice boxes. Each pretzel bag costs p dollars and each juice box costs 3 dollars.

Grouped form

$$2(p+3)$$

Shows two identical snack sets.

Expanded form

$$2p+6$$

Shows pretzel cost and juice cost separately.

1. Explain why the expressions are equivalent.
2. Which form best shows the story?
3. Which form would be easier to evaluate if $p=4$?

You Try It 4

A party favor bag has 5 stickers and 5 erasers. Each sticker costs s cents and each eraser costs 8 cents.

1. Write a grouped expression.
2. Write an expanded expression.
3. Explain what each form helps you see.

Stop and Discuss

1. Why might a teacher ask for an explanation instead of only a simplified answer?
2. When might the unsimplified form be more useful than the simplified form?

Summary

- An expression is a math phrase without an equals sign.
- Structure means how an expression is built.
- Terms are separated by addition or subtraction.
- Like terms have matching variable parts and can be combined.
- The distributive property rewrites a grouped expression into an equivalent expanded form.
- Equivalent expressions may look different but have the same value for every input.
- Different forms can reveal different information about the same situation.

Common Mistakes

- **Combining unlike terms:** $3x+4$ cannot become $7x$ because $3x$ and 4 are different kinds of parts.
- **Dropping a negative sign:** In $6x-2+3x$, the -2 stays negative when terms are rearranged.
- **Distributing to only one term:** $4(x+6)$ becomes $4x+24$, not $4x+6$.
- **Thinking equivalent means identical-looking:** Equivalent expressions can look different and still always have the same value.
- **Simplifying without meaning:** A simplified form is useful, but students should still know what the parts represent.
- **Choosing one form for every situation:** Different forms can be helpful for different reasons.

Optional Future Task Reflection

Optional Reflection: Look back at the future task. Do not solve it yet.

1. Highlight one expression card that makes more sense now than it did at the beginning.
2. Circle one expression form that still feels unfamiliar.
3. Write one sentence about which form you would look at first if you needed to explain the structure.